INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME			
CLASS		INDEX NUMBER	

BIOLOGY 9648/01

Paper 1 Multiple Choice **20 September 2011**

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

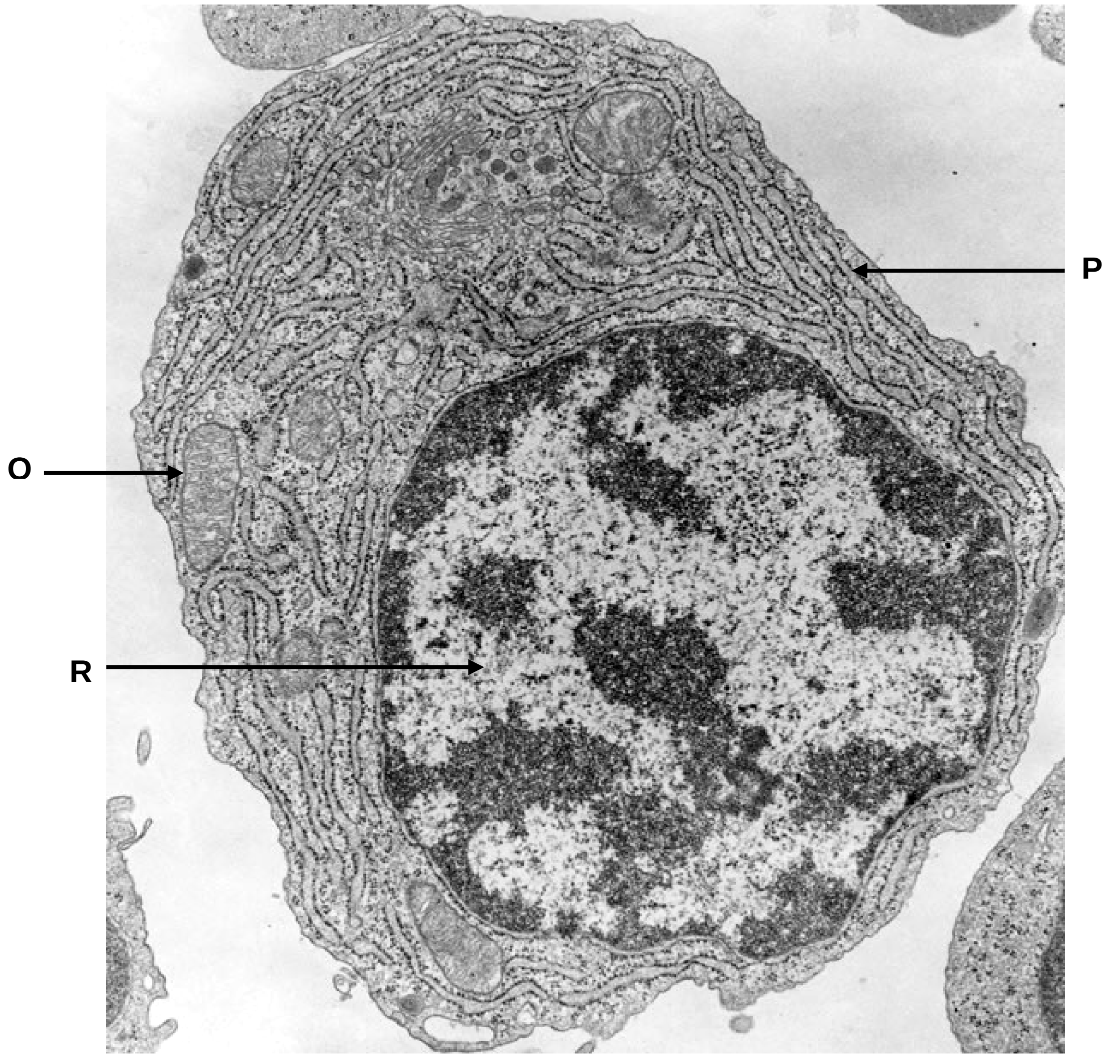
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **22** printed pages.

1 The figure below shows an electron micrograph of a cell.



Which of the following about structures **P**, **Q** and **R** are correct?

	P	Q	R
A	provides large surface area for attachment of ribosomes	modification of transcripts	active condensation of chromatin
B	transport of proteins to Golgi apparatus	formation of ATP from light energy	active transcription of genes
C	synthesis and processing of membrane proteins	sequestering of protons within a specialised compartment	active transcription of genes
D	prevents proteins from degradation by cytoplasmic enzymes	allows the oxidation of pyruvate	active replication of DNA

2 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

A Presence or absence of 180° rotation of alternate subunits.

B Presence or absence of branching in the macromolecule

C Subunits are either monosaccharides or amino acids.

D Different tendency of macromolecule to form hydrogen bonds with water.

3 The **R** group of the amino acid *serine* is $-CH_2-OH$. The **R** group of the amino acid *alanine* is $-CH_3$. Where would you expect to find these amino acids in a globular protein in aqueous solution?

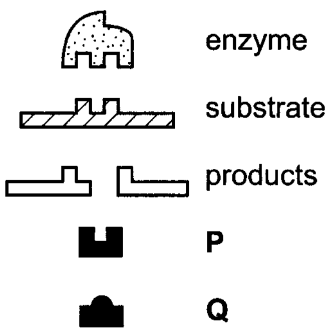
A Serine would be in the interior, and alanine would be on the exterior of the globular protein.

B Alanine would be in the interior, and serine would be on the exterior of the globular protein.

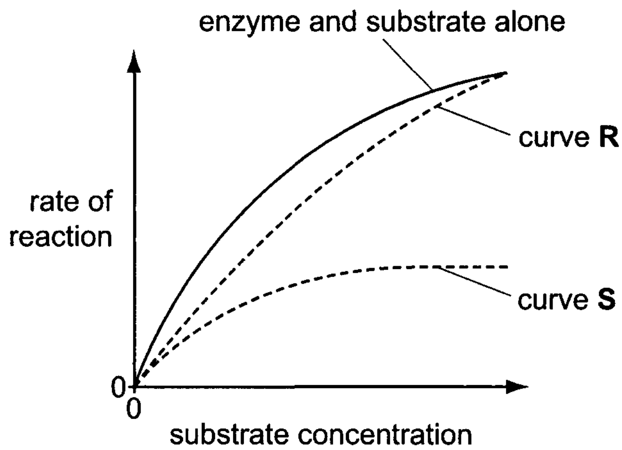
C Both serine and alanine would be in the interior of the globular protein.

D Both serine and alanine would be in the interior and on the exterior of the globular protein.

4 The diagram shows an enzyme molecule with its normal substrate and products. **P** and **Q** are other molecules that can bind to the enzyme.



The graph shows the effect of **P** and **Q** on the rate of reaction of the enzyme at different substrate concentrations.



Which statement correctly describes the activity of the enzyme?

A **P** is a competitive inhibitor which binds to the active site, resulting in curve **R**.

B **P** is a non-competitive inhibitor which distorts the shape of the enzyme, resulting in curve **S**.

C **Q** is a competitive inhibitor which distorts the shape of the enzyme, resulting in curve **R**.

D **Q** is a non-competitive inhibitor which binds to the active site, resulting in curve **S**.

5 In the production of isoleucine from threonine (**pathway P**), the end product acts as an inhibitor of the first enzyme in the pathway. In the production of arginine (**pathway Q**), the end product has no influence on other enzymes in the pathway. The two pathways are shown in the figure.

Pathway P:

inhibits enzyme 1

$$\text{threonine} \xrightarrow[\text{enzyme 1}]{} \text{compound X} \xrightarrow[\text{enzyme 2}]{} \text{compound Y} \xrightarrow[\text{enzyme 3}]{} \text{isoleucine}$$

Pathway Q:

$$\text{substrate} \xrightarrow[\text{enzyme 1}]{} \text{compound M} \xrightarrow[\text{enzyme 2}]{} \text{compound N} \xrightarrow[\text{enzyme 3}]{} \text{arginine}$$

Which of the following conclusions is **correct**?

I	In **pathway P**, if the production of enzyme 3 stops, there would be increased production of isoleucine.
II	In **pathway Q**, if the production of enzyme 3 stops, there would be decreased production of arginine.
III	In **pathway P**, provided all enzymes are present, the production of isoleucine would be continuous if there was a continuous supply of threonine.
IV	In **pathway Q**, provided all enzymes are present, the production of arginine would be continuous if there was a continuous supply of substrate.

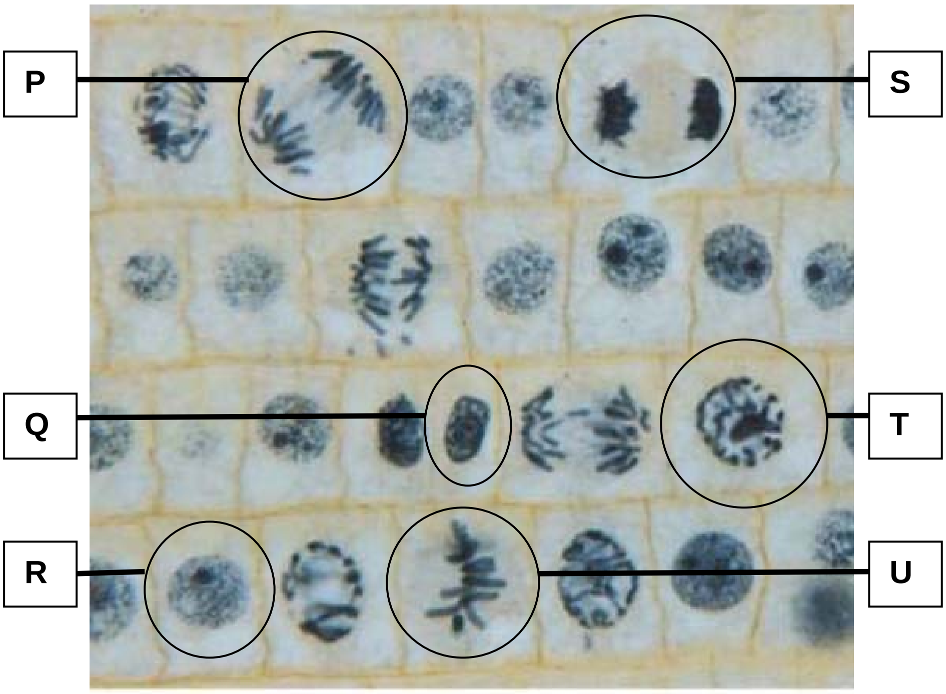
A **I** and **III** only

B **I** and **IV** only

C **II** and **III** only

D **II** and **IV** only

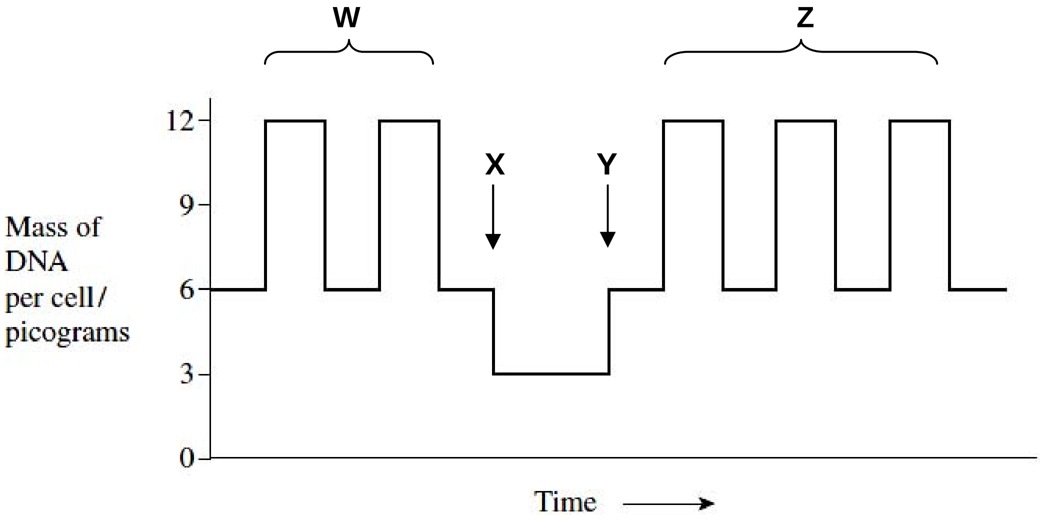
6 The photomicrograph below show a section of the onion root tip. The cells are at different stages of mitosis.



Which of the following shows the correct sequence of events that occurs in these cells?

	First					Last
A	**R**	**T**	**U**	**P**	**S**	**Q**
B	**T**	**U**	**P**	**S**	**Q**	**R**
C	**Q**	**T**	**P**	**S**	**U**	**R**
D	**T**	**R**	**P**	**S**	**U**	**Q**

7 The graph shows the mass of DNA present per human cell during sperm production, fertilization and early development of the embryo.



Which of the following statements about the graph is / are **true**?

I The cells formed during **W** and **Z** are genetically identical to one another as a result of mitosis.

II The cell plate starts to reform during **X**.

III DNA replication occurs during **Y**.

IV Formation of totipotent cells occurs during **Z**.

A **IV** only

B **I** and **II** only

C **I** and **III** only

D **II** and **IV** only

8 During gametogenesis, non-disjunction occurs in one daughter cell during meiosis II. What would be the result at the completion of meiosis?

A All gametes would be haploid.

B Half of the gametes would be haploid and the other half would be diploid.

C Half of the gametes would have n chromosomes, one quarter would have (n + 1) chromosomes and another quarter would have (n - 1) chromosomes

D Half of the gametes would have (n + 1) chromosomes and the other half would have (n - 1) chromosomes.

9 Human insulin is a protein containing 51 amino acids. These include 17 of the 20 different amino acids commonly occurring in proteins.

What is the minimum number of different kinds of amino-acyl tRNA synthetase molecules involved in the synthesis of insulin?

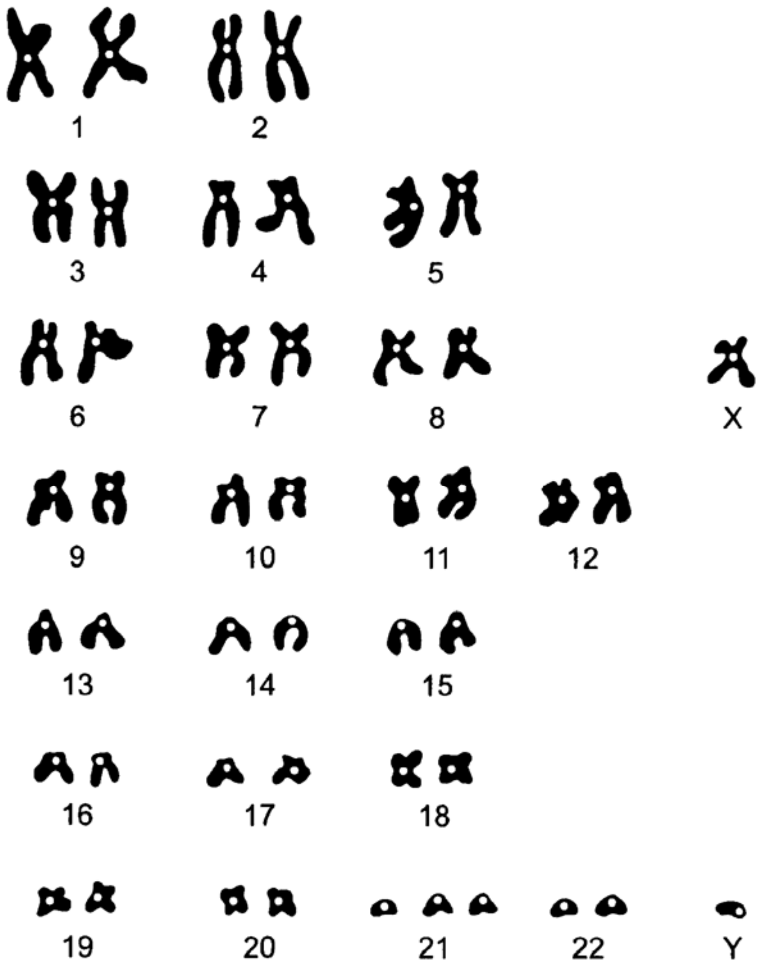
A 3

B 17

C 20

D 51

10 The diagram shows a human karyotype.



The person who has this karyotype is a male with

A a gene mutation.

B aneuploidy.

C normal karyotype.

D polyploidy.

11 The following shows part of a normal DNA sequence of a gene and the first 5 amino acids it codes for. A mutation occurred resulting in a different polypeptide sequence.

Normal DNA sequence : G A T C A T C A T A T G A G C

Normal polypeptide sequence : Leu – Val – Val – Tyr – Ser

Mutant polypeptide sequence : Leu – Val – Ile

The table shows 6 amino acids and their corresponding codons.

Amino Acid	*mRNA Codon*
Ile	AUA
Gly	GGA
Leu	CUA / CUC
Met	AUG
Ser	UCG
Thr	ACU
Tyr	UAC / UAU
Val	GUA / GUC
STOP	UAA / UAG / UGA

Which of the following mutations is most likely to have occurred?

A Deletion at base 7

B Deletion at base 7 and 8

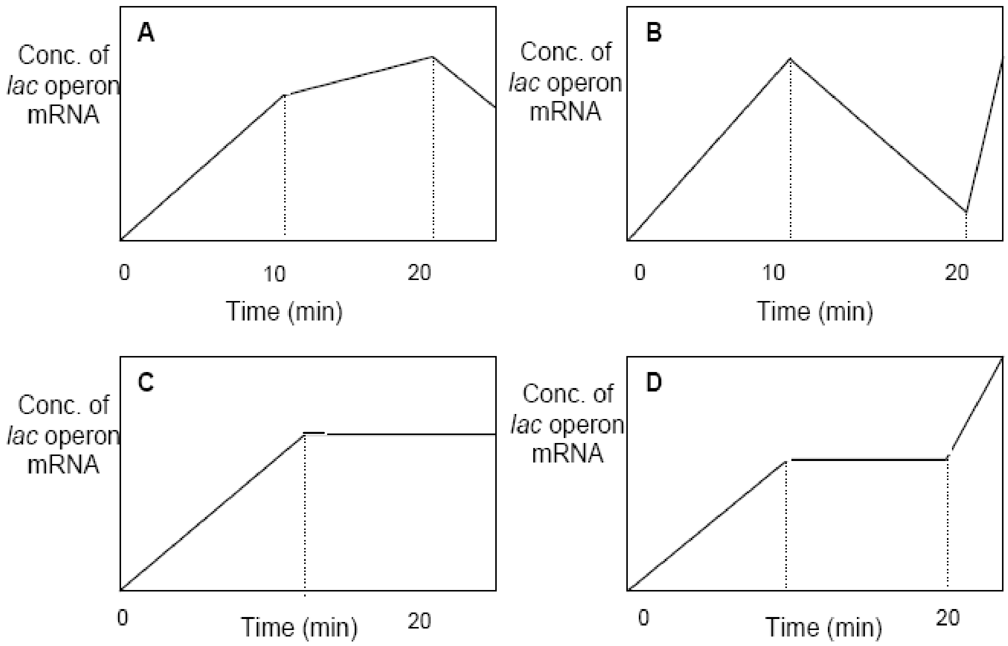
C Inversion from base 6 to 12

D Substitution at base 7

12 IPTG is an analogue of lactose that binds to the *lac* repressor in the same fashion as allolactose. However, it cannot be metabolised by β-galactosidase.

E. coli cells, which were grown in the absence of lactose and glucose, were initially supplemented with IPTG. After 10 minutes, glucose was added to the cells. 10 minutes later, cAMP was added to the cells.

Which of the following graphs represents the amount of *lac* operon mRNA during the time course of the experiment?



13 If two populations of genetically different bacteria cultured in a U-shaped tube are separated by a membrane filter (which does not allow phage particles to pass), but the recombination takes place anyway.

What is the mechanism of the genetic exchange?

A specialized transduction

B generalized transduction

C transformation

D conjugation

14 In *Escherichia coli*, the synthesis of tryptophan is controlled by the tryptophan operon that is repressed in the presence of excessive tryptophan.

When a mutant strain that has lost the regulatory gene of the tryptophan operon is placed in a medium that contains all nutrients the cells need to grow except tryptophan, which of the following will occur?

A The cells will grow until excessive tryptophan arrests the expression of the operon.

B The cells will continue to grow even though excess tryptophan is synthesized.

C The cells will never grow unless tryptophan is added to the medium.

D The cells will not grow until enough tryptophan has been synthesized to activate the corepressor.

15 How do viruses cause disease in animals?

1 They inhibit normal host cell DNA, RNA, or protein synthesis.

2 They degrade the host cell's chromosomes.

3 They disrupt the oncogenes of the host cell causing uncontrolled cell division.

4 Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defenses.

5 They deplete the host cell of cellular materials essential for metabolic functions.

A 1, 2 and 5

B 1, 2, 3 and 5

C 1, 2, 4 and 5

D All of the above

16 Which of the following statements about the life cycle of the Human Immunodeficiency Virus are **false**?

1 The HIV particles recognise the host cell through the sialic-acid containing proteins or lipids on the membrane of the host cell.

2 The genome of HIV consists of 2 copies of linear, single-stranded RNA enclosed in a icosahedral capsid.

3 The viral DNA which enters the host cell's nucleus will be integrated into the genetic material of the host cell using the host cell's enzyme, integrase.

4 The viruses are released from the host cell by exocytosis.

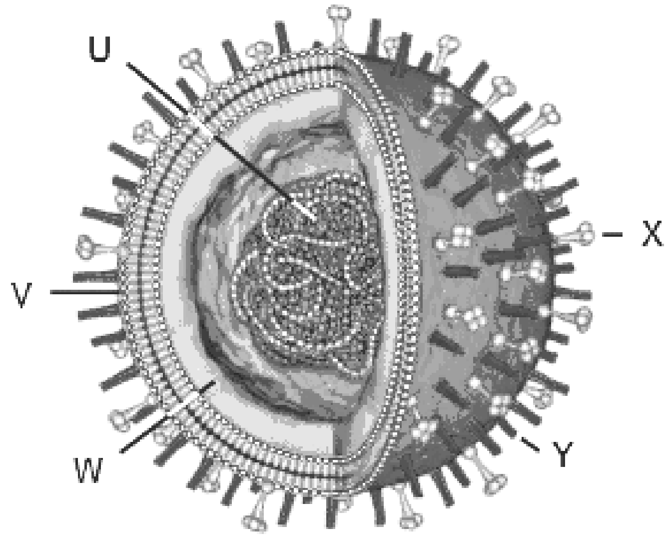
A 1 and 3

B 2 and 4

C 1, 3 and 4

D All of the above

17 The figure below represents the structure of the influenza virus.



Which combination correctly identifies the functions of the lettered components?

	U	W	X	Y
A	serves as template for synthesis of complementary RNA strand	protects the viral genetic material	release of newly produced viral particles from the host cell	attaches virus to target cells
B	serves as template for synthesis of complementary DNA strand	protects the viral genetic material	release of newly produced viral particles from the host cell	attaches virus to target cells
C	serves as template for synthesis of complementary RNA strand	fuses with membrane of target cells	protects the viral genetic material	release of newly produced viral particles from the host cell
D	serves as template for synthesis of complementary DNA strand	fuses with membrane of target cells	attaches virus to target cells	release of newly produced viral particles from the host cell

18 Which of the following describes common features of the telomere and the centromere?

A Both areas have proteins associated with them.

B Both are made up of DNA with a high incidence of adenine and thymine.

C Both DNA sequences are conserved throughout the life of the organism.

D During interphase, DNA replication takes place simultaneously in both regions.

19 Dioxin, produced as a by-product of various chemical processes, is suspected of causing cancer and birth defects in animals and humans. It acts by entering cells and binding to activate proteins, altering the pattern of gene expression of proto-oncogenes.

Which of the following is most likely to be directly affected by dioxin?

A Enhancers

B Histone deacetylase

C Repressors

D Transcription factors

20 Four different genes are regulated in different ways.

Gene 1 undergoes tissue-specific patterns of alternative splicing.

Gene 2 is part of a group of structural genes controlled by the same regulatory sequences.

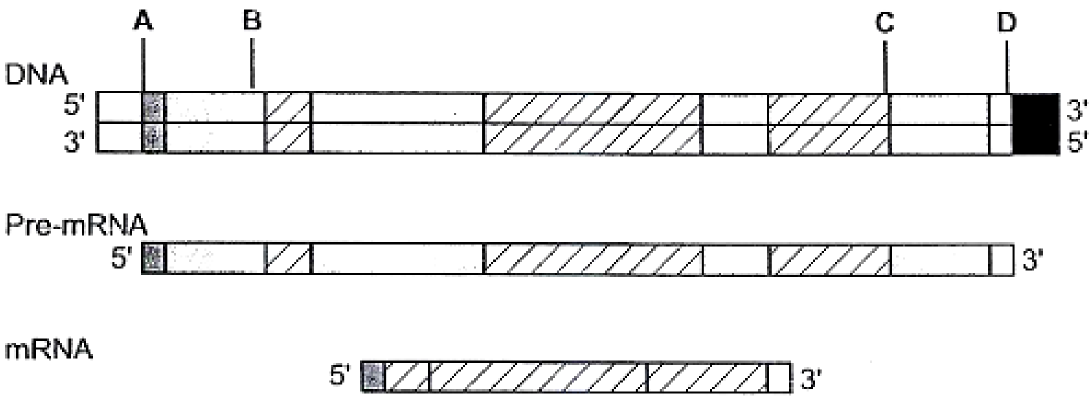
Gene 3 is in some circumstances subject to methylation.

Gene 4 codes for a repressor protein which acts at an operator site close by.

Which row of the table correctly identifies which genes are prokaryotic and which are eukaryotic?

	prokaryotic	eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

21 The diagram shows a gene, its pre-mRNA and mature mRNA after processing.



Which region when mutated is most likely to result in error during post-transcriptional splicing?

22 Stem cells have active telomerase that prevents chromosome shortening with every DNA replication. Which of the following describes the role of telomerase?

A It acts as a buffer to prevent erosion of subtelomeric genes.

B It extends the parental DNA strand at the 3' end.

C It lengthens the daughter DNA strand at the end of replication.

D It prevents the end replication problem from occurring.

23 In a species of mammal, chromosome 7 contains genes for two enzymes.

Enzyme **P** catalyses an early step in a metabolic pathway, and enzyme **Q** catalyses a later step in the same pathway. Both genes have two alleles, so there are nine possible genotypes.

Which terms appropriately describe this situation?

A dihybrid inheritance and linkage only

B epistasis and multiple alleles only

C linkage, dihybrid inheritance and epistasis

D multiple alleles, dihybrid inheritance and linkage

24 Two pairs of allele govern the colour of onion bulbs. A pure-breed red strain crossed with a pure-breed white strain produces all red F1.

Self crossing of F1 produced 47 white, 38 yellow and 109 red bulbs. What does this experiment demonstrate?

A Codominance

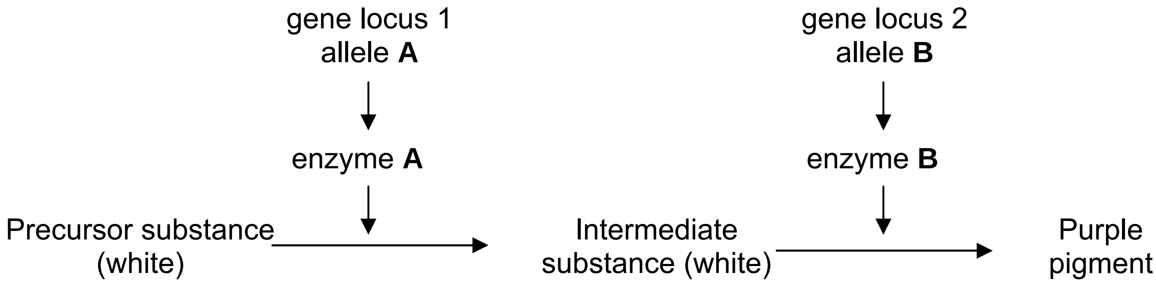
B Dominant epistasis

C Recessive epistasis

D There is not enough evidence to come to a conclusion

25 The inheritance of colour in sweat peas is controlled by two pairs of alleles (**A** and **a**, and **B** and **b**) at two different gene loci.

The diagram shows how the dominant alleles at these two gene loci determine flower colour by controlling the synthesis of a purple pigment.



If a double heterozygote plant (**AaBb**) is *self-pollinated*, what is the number of phenotypes observed in the resulting offspring?

A 1

B 2

C 3

D 4

26 In one of the experiments to test the 3:1 ratio of Mendel's laws, 705 plants are observed to have red flowers and 224 had white flowers.

The **P** (probability) value, associated with the calculated chi-square value of 0.400, is more than 0.10, under the 95% (P=0.05) confidence level. The **P** refers to

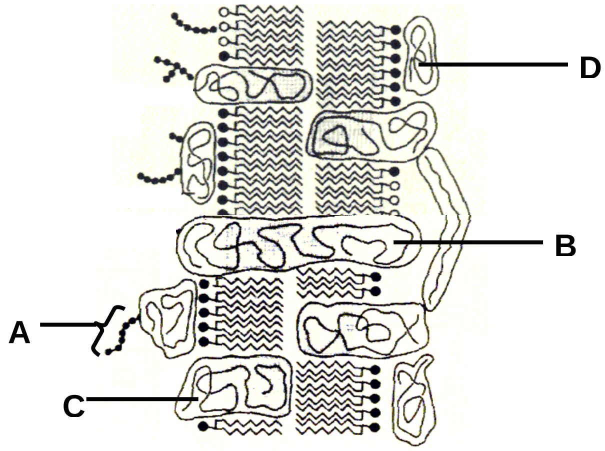
A the chance that there is a 10% probability that any deviation from the expected result is due to a random event only.

B the chance that there is a 10% probability that any deviation from the expected result is significant.

C the chance that there is no difference between the expected and observed results.

D the chance that there is a difference between the expected and observed results.

27 The diagram below shows a section of a cell surface membrane.



Which of the following statements are correct?

I Structure **A** is found only on the extracellular surface of the membrane.

II Structure **B** may contain a channel that is hydrophobic to allow ions to move across the membrane.

III Structure **B** is able to bind to certain substances and change its conformation to transport these substances into the cell.

IV Structure **C** is able to change the structure of substances such that the substance can now diffuse across the membrane

V Structure **D** is found only on one surface of the membrane.

A **I, II, III** and **IV**

B **II, III** and **V**

C **I** and **III** only

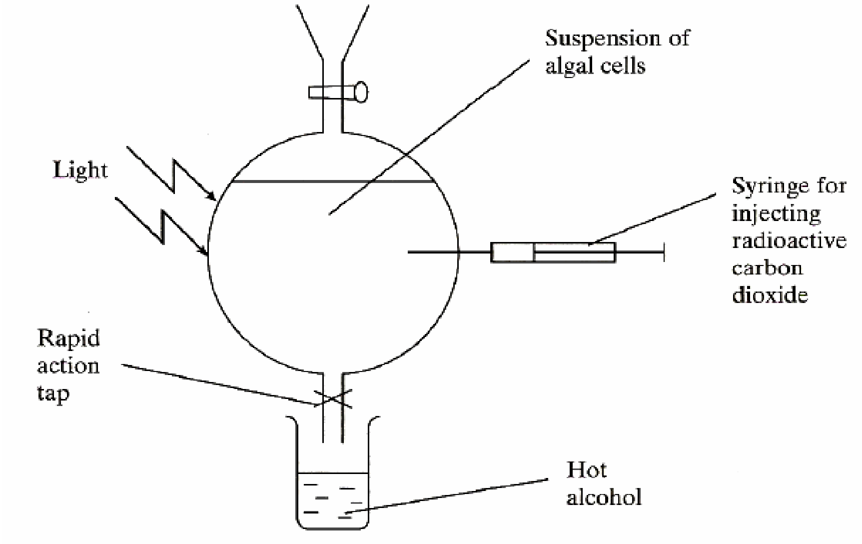
D **I** and **V** only

28 Which of the following is correct in the hormonal control of blood glucose level?

	Hormone	Receptor which hormone binds to	Effect
A	insulin	tyrosine kinase receptor	increase glycogenesis
B	insulin	G protein-linked receptor	increase gluconeogenesis
C	glucagon	tyrosine kinase receptor	increase gluconeogenesis
D	glucagon	G protein-linked receptor	increase glycogenesis

29 An investigation was carried out to find out the sequence of biochemical changes that occur during photosynthesis.

Radioactive carbon dioxide was added to a suspension of algal cells, and they were allowed to photosynthesise. At intervals, samples of the suspension were removed into hot alcohol. These samples were analysed for different radioactively labelled compounds. The experimental set up is shown below.



Samples were removed from the suspension at five different times, between 5 seconds and 600 seconds after the start of the experiment. In each sample, the amount of radioactivity present in four different organic compounds, **P**, **Q**, **R** and **S** was measured. The table shows the results.

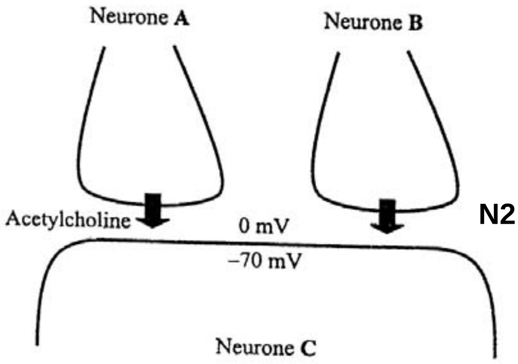
organic compound	amount of radioactivity present / arbitrary units				
	5 s	15 s	60 s	180 s	600 s
P	0.01	0.02	0.08	0.17	0.67
Q	1.00	2.00	3.10	3.15	3.15
R	0.10	1.50	2.20	2.30	2.40
S	0.05	0.11	0.16	1.00	1.00

What are the most likely identities of the four organic compounds?

	P	**Q**	**R**	**S**
A	TP (GALP)	GP	glycerate 1,3-bisphosphate	RuBP
B	GP	RuBP	TP (GALP)	glycerate 1,3-bisphosphate
C	RuBP	GP	glycerate 1,3-bisphosphate	TP (GALP)
D	TP (GALP)	RuBP	GP	glycerate 1,3-bisphosphate

30 The diagram below shows axon terminals from each of two neurones, **A** and **B**, forming synapses with a third neurone, **C**.

When stimulated, neurone **A** releases acetylcholine while neurone **B** releases a structurally different neurotransmitter, N2. Concentrations of some ions are also shown.



	Ion concentration / mmol dm^{-3}		
	Na^+	**K^+**	**Cl^-**
Outside neurone C	145	12	120
Inside neurone C	12	155	5

Which of the following statements best explains why an action potential is not triggered in neurone **C** when both neurones **A** and **B** are stimulated?

A N2 competes with acetylcholine for receptor binding at the postsynaptic membrane of neurone **C**, resulting in less Na^+ ions entering to cause depolarisation.

B N2 causes the membrane potential of neurone **C** to be less negative, therefore more acetylcholine must be released by neurone **A** to cause greater influx of Na^+ ions.

C N2 increase the permeability of the postsynaptic membrane of neurone C to Cl^- ions, resulting in hyperpolarisation.

D Acetylcholine is released slowly from neurone **A** while N2 is released rapidly from neurone **B**.

31 Three proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

	Protein A	Protein B	Protein C
ATPase activity	+	-	-
GTPase activity	-	-	+
DNA binding	-	+	-
Transmembrane Domain	+	-	+

Key: (+) = present, (-) = absent

Which of the following correctly shows the identity of these three proteins?

	Protein A	Protein B	Protein C
A	Na^+- K^+ pump	Ras protein	Glucagon receptor
B	Glucagon receptor	Ras protein	Na^+- K^+ pump
C	Na^+- K^+ pump	Glucagon receptor	Ras protein
D	Ras protein	Glucagon receptor	Na^+- K^+ pump

32 Which statements are true?

1 Continuous variation is always influenced by the environment.
2 Continuous variation is always caused by many genes.
3 Discontinuous variation is always caused by one gene.
4 Discontinuous variation is sometimes influenced by the environment.

A 1 and 4
B 2 and 3
C 2 and 4
D 3 and 4

33 The α, β and γ globin chains of human, chimpanzee, gorilla and gibbon haemoglobin were analysed. The number of differences in the amino acid sequences in comparison with the human molecules is shown in the table.

species	number of differences in amino acid sequence from human molecules		
	α globin	β globin	γ globin
chimpanzee	0	0	1
gorilla	1	1	1
gibbon	3	3	2

Which of the following is a most likely conclusion?

A Gibbons evolved into gorillas which then evolved into chimpanzees and subsequently into humans.
B Humans and gibbons shared a more recent common ancestor than humans and chimpanzees.
C Humans and gorillas shared a more recent common ancestor than humans and gibbons.
D Humans and gibbons do not share a common ancestor.

34 The incidence of infections with *Staphylococcus aureus* resistant to the antibiotic methicillin is rising yearly.

What is a possible evolutionary reason for this?

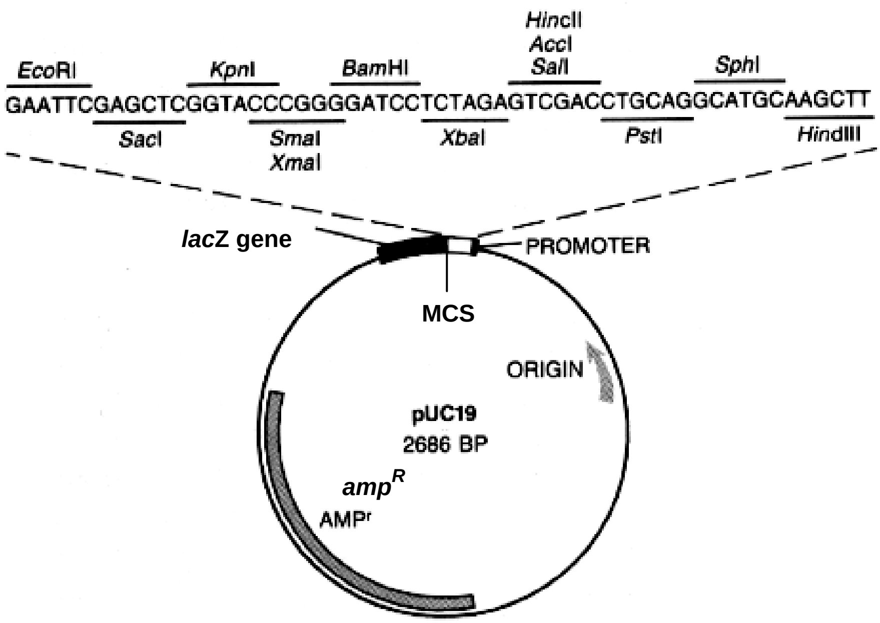
A Artificial selection of resistant *S. aureus* in response to methicillin use.

B Heterozygote advantage for reduced immunity in the human population.

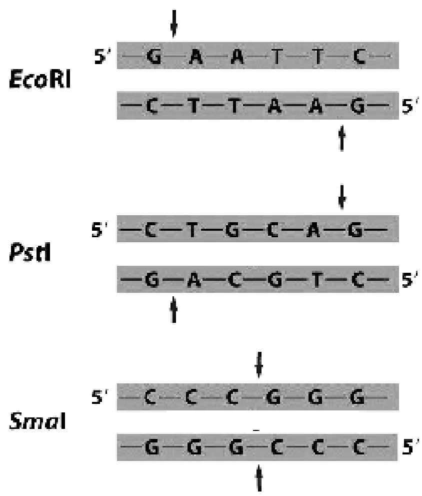
C Methicillin acts as selection pressure causing resistant genes to mutate.

D Natural selection of resistant *S. aureus* in response to methicillin use

35 The figure shows the plasmid map of pUC19, which includes the positions of the origin of replication (ORI), multiple cloning sites (MCS) and two genes - *amp*^{R} and *lac*Z. pUC19 is used for the production of human anti-thrombin III (hATIII) in the bacterium *E. coli*.



The restriction sites of three particular restriction enzymes, *Eco*RI, *Pst*I and *Sma*I are shown below.



[Turn over

Which of the following can be concluded from the figures?

I	Any restriction enzyme whose restriction site is found within the MCS can be used to cleave pUC19.
II	pUC19 cleaved with *Eco*RI can anneal to hATIII cDNA cleaved with *Pst*I as both restriction enzymes generate single-stranded overhangs of the same length.
III	The use of *Sma*I will require an additional step of ligating adaptors to the ends of pUC19 and hATIII cDNA.
IV	Colonies of *E. coli* transformed with recombinant pUC19 will survive on agar plates containing ampicillin and X-gal, giving a blue appearance.

A **I** and **III** only

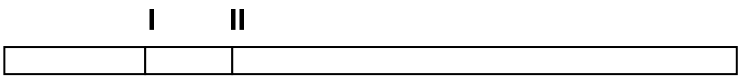
B **I** and **IV** only

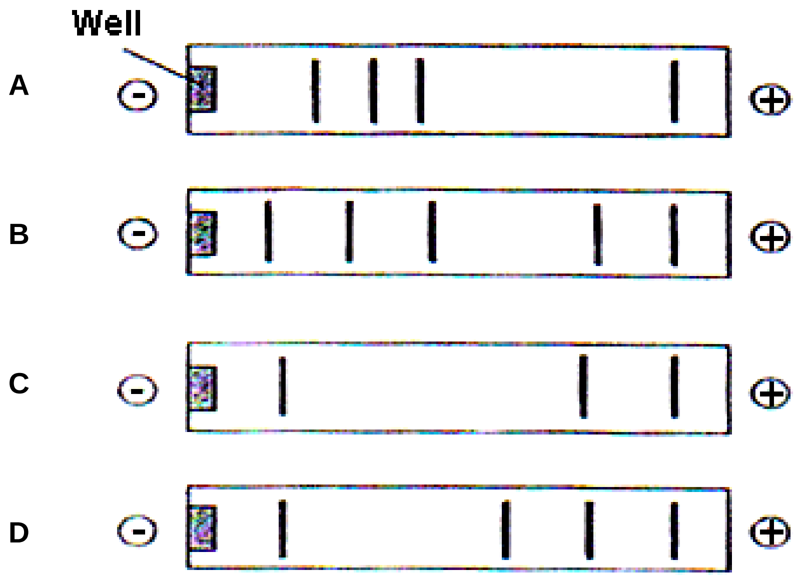
C **II** and **III** only

D All of the above

36 The restriction fragment shown below contains a gene whose recessive allele is lethal. The normal allele has restriction sites for restriction enzyme *PstI* at sites **I** and **II**. The disease-causing allele lacks restriction site **I**. An individual who had a sister with the lethal trait is being tested to determine if he is a carrier of that lethal trait.

Which of the band patterns would be produced on a gel if he is a carrier?





37 Which of the following possess the greatest limitation of the Human Genome Project in disease analysis?

A It is not possible to store the genomic information of all individuals.

B There is too large a genetic difference between individuals to determine the loci of genes of interest.

C Too few model organisms exist for comparative genomic studies to take place.

D Environmental factors and gene interactions in disease are not taken into account.

38 Stem cells can be divided into four main types. Which of the four types of stem cell can differentiate into a limited range of tissues?

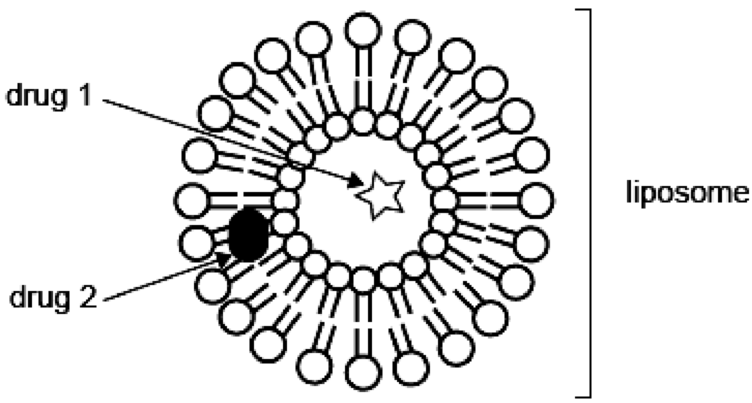
A Adult stem cells

B Embryonic stem cells

C Foetal stem cells

D Umbilical stem cells

39 Liposomes can be found in the cytosol and are formed from a double layer of phospholipid molecules identical to those found in plasma membranes. Liposomes can also be manufactured and used to carry medicinal drugs into cells. The diagram below shows the association of two drugs with a liposome.



Which of the following conclusions can be drawn from the above diagram?

A Drug 1 is hydrophobic.

B Drug 2 is made of lipids.

C Both drugs are amphoteric.

D The interior of the liposome is aqueous.

40 Which of these processes could lead to an increase in crop yield?

P inserting genes for vitamin A production into rice

Q inserting genes for Bt toxin into cotton

R inserting genes for herbicide resistance into oilseed rape

S inserting genes for anti-freeze proteins into soya beans

A P and Q

B Q and R

C R and S

D S and P

End of Paper

2010 JC2 Prelim Exam 2
Paper 1

Qn	Answer	Qn	Answer	Qn	Answer	Qn	Answer
1	**D**	11	**C**	21	**B**	31	**A**
2	**A**	12	**B**	22	**B**	32	**C**
3	**B**	13	**C**	23	**C**	33	**C**
4	**A**	14	**B**	24	**C**	34	**D**
5	**D**	15	**A**	25	**B**	35	**A**
6	**A**	16	**C**	26	**A**	36	**B**
7	**A**	17	**A**	27	**C**	37	**D**
8	**C**	18	**A**	28	**A**	38	**A**
9	**B**	19	**D**	29	**C**	39	**D**
10	**B**	20	**D**	30	**C**	40	**B**